

Spaceship in the atmosphere — Solution

X25 (2025) — English translation

A1

0.20 pts

First, we obtain the exact equations that describe the dynamics of the spacecraft. Express the time derivatives of altitude h and range s in terms of v , γ , r_0 , r .

The rate of decrease in height is the vertical projection of speed $v \sin \gamma$, the speed of horizontal movement $v \cos \gamma$ after projection onto the surface of the Earth decreases by r/r_0 times.

Answer: \textbf{Answer:}

$$\dot{h} = -v \sin \gamma, \quad \dot{s} = v \cos \gamma \cdot r_0/r$$

A2

1.00 pts

Find the derivatives of the velocity modulus v and angle γ . The answer may also include the lift force L , the air resistance force D , the acceleration due to gravity at a given height g , the mass of the spacecraft m , v , γ , r .

Newton's second law in projection onto the axes along and perpendicular to the speed:

$$ma_\tau = -D + mg \sin \gamma, \quad ma_n = -L + mg \cos \gamma.$$

Tangential acceleration $a_\tau = \dot{v}$, therefore

$$a_\tau = -\frac{D}{m} + g \sin \gamma.$$

Normal acceleration $a_n = \omega v$, where ω - angular velocity поворота вектора скорости, которую считаем положительной, if angular velocity поворачивается вправо. Эта velocity не equals производной угла γ , since направление горизонтали меняется от точки к точке с угловой velocity $\omega_h = v \cos \gamma/r$, then

$$\omega = \dot{\gamma} + \frac{v \cos \gamma}{r},$$

therefore equation for нормального ускорения

$$mv \left(\dot{\gamma} + \frac{v \cos \gamma}{r} \right) = -L + mg \cos \gamma,$$

where do we find the required derivative

Answer: \textbf{Answer:}

$$\dot{v} = -\frac{D}{m} + g \sin \gamma, \quad \dot{\gamma} = -\frac{L}{mv} + \frac{g}{v} \cos \gamma - \frac{v \cos \gamma}{r}$$

B1

0.40 pts

Obtain an expression for the derivative dv/dh . Represent the answer as

$$\frac{dv}{dh} = Af(v, h),$$

where A - постоянная, которая может зависеть от всех постоянных, характеризующих

космический корабль and атмосферу ($m, S, C_D, C_L, \rho_0, \beta, \gamma$), а f – некоторая function высоты and скорости. В дальнейшем используйте постоянную A to write answers.

Let us substitute the expression for the air resistance force D into the equation of motion and obtain, neglecting gravitational acceleration

$$\dot{v} = -\frac{\rho_0 S C_D}{2m} v^2 e^{-\beta h}.$$

Taking into account формулы for $\dot{h} = -v \sin \gamma$ we get

$$\frac{dv}{dh} = \frac{\dot{v}}{\dot{h}} = \frac{\rho_0 C_D S}{2m \sin \gamma} \cdot v e^{-\beta h}$$

Answer: \textbf{Answer:}

$$\frac{dv}{dh} = A v e^{-\beta h}, \quad A = \frac{\rho_0 C_D S}{2m \sin \gamma}$$

B2

0.80 pts

Find the dependence of the speed of the spacecraft on altitude. At the initial height h_0 the speed was equal to v_0 . Express your answer in terms of A, β, h, v_0, h_0 . Also get an approximate answer, assuming that at an altitude of h_0 the atmospheric density is negligible (in this case the answer should not include h_0). In the following, use an approximate expression throughout.

Dividing the variables, we get

$$\frac{dv}{v} = A e^{-\beta h} dh,$$

проинтегрировав, we get

$$\int_{v_0}^v \frac{dv}{v} = \int_{h_0}^h A e^{-\beta h} dh, \quad \ln \frac{v}{v_0} = -\frac{A}{\beta} (e^{-\beta h} - e^{-\beta h_0}).$$

Hence let us find expression for скорости

$$v = v_0 \exp \left(-\frac{A}{\beta} (e^{-\beta h} - e^{-\beta h_0}) \right)$$

For/At достаточно большом значении h_0 terms $e^{-\beta h_0} \ll e^{-\beta h}$ можно пренебречь

$$v = v_0 \exp \left(-\frac{A}{\beta} e^{-\beta h} \right)$$

Answer: \textbf{Answer:}

$$v = v_0 \exp \left(-\frac{A}{\beta} (e^{-\beta h} - e^{-\beta h_0}) \right), \quad v \approx v_0 \exp \left(-\frac{A}{\beta} e^{-\beta h} \right)$$

B3

0.30 pts

Find the dependence of the acceleration a (that is, the derivative of the velocity modulus) of the spacecraft on altitude. Express your answer in terms of v_0, A, β, h, γ .

Let us substitute the formula for velocity into the equation for tangential acceleration and obtain

$$a = \dot{v} = -\frac{D}{m} = -\frac{\rho_0 S C_D}{2m} v^2 e^{-\beta h} = -A v_0^2 \sin \gamma \exp \left(-\frac{2A}{\beta} e^{-\beta h} - \beta h \right)$$

Answer: \textbf{Answer:}

$$a = -Av_0^2 \sin \gamma \exp \left(-\frac{2A}{\beta} e^{-\beta h} - \beta h \right)$$

B4

0.50 pts

Find the height h_c at which the acceleration modulus is maximum. Obtain a formula for the maximum value of acceleration modulus a_{max} and velocity v at height h . Express your answers using A, β, γ, v_0 .

The acceleration module is maximum, with a minimum exponent,

$$\frac{d}{dh} \left(-\frac{2A}{\beta} e^{-\beta h} - \beta h \right) = 2Ae^{-\beta h} - \beta = 0.$$

Hence критическое value высоты

$$h_c = \frac{1}{\beta} \ln \frac{2A}{\beta}.$$

Substituting это value высоты в формулы for высоты and ускорения, we get

$$a_{max} = Av_0^2 \sin \gamma \exp \left(-\frac{2A}{\beta} e^{-\beta h_c} - \beta h_c \right) = \frac{\beta v_0^2 \sin \gamma}{2e}$$

$$v_c = v_0 \exp \left(-\frac{A}{\beta} e^{-\beta h_c} \right) = v_0 e^{-1/2}$$

Answer: \textbf{Answer:}

$$h_c = \frac{1}{\beta} \ln \frac{2A}{\beta}, \quad v_c = v_0 e^{-1/2}, \quad a_{max} = \frac{\beta v_0^2 \sin \gamma}{2e}$$

B5

0.60 pts

For the given numerical data, find the speed (in km/s) and acceleration (in units of g_0) at heights $h_1 = 80$ km, $h_2 = 60$ km, $h_3 = 40$ km.

Substituting the height values into the formulas for speed and acceleration, we get

Answer: \textbf{Answer:} h , km a/g_0 v , km/ 80 -0.20 7.7 60 -2.8 7.2 40 -5.9 2.7

B6

0.20 pts

Find the maximum acceleration (in units of g_0) and the critical height h_c at which it is achieved.

Answer: \textbf{Answer:}

$$a_{max} = 8.6 g_0, \quad h_c = 45.5 \text{ km}$$

C1

0.10 pts

Find the first escape velocity v_s - the speed of the spacecraft in a circular orbit, the radius of which is equal to the radius of the Earth. Express your answer in terms of r_0, g_0 .

Let us write the equation of motion for acceleration when moving in a circle

$$\frac{v^2}{r_0} = g_0,$$

Answer: \textbf{Answer:}

$$v_s = \sqrt{g_0 r_0}$$

C2

0.40 pts

Determine the speed at which the spacecraft must move at altitude h for the motion described in the introduction to this part. Express your answer in terms of v_s , ρ_0 , β , C_L , S , m , r_0 , h .

Let us write the equation for normal acceleration, taking into account that the derivative $\dot{\gamma}$ is small, and $\cos \gamma \approx 1$ and it can be neglected

$$-\frac{L}{m} + g_0 - \frac{v^2}{r_0} = 0.$$

Substituting формулу for подъемной силы, a also for зависимости плотности воздуха от высоты, we get

$$v^2 \left(\frac{1}{r_0} + \frac{C_L S \rho_0}{2m} e^{-\beta h} \right) = g_0,$$

hence let us find

$$v^2 = \frac{g_0 r_0}{1 + \frac{C_L \rho_0 r_0 S}{2m} \cdot e^{-\beta h}}$$

Answer: \textbf{Answer:}

$$v = \frac{v_s}{\sqrt{1 + \frac{C_L \rho_0 r_0 S}{2m} \cdot e^{-\beta h}}}$$

C3

0.40 pts

Determine the acceleration a (derivative of the velocity modulus) created by the force of air resistance during such movement. Express your answer in terms of v , C_L , C_D , r_0 , g_0 .

Lift and air resistance are related by the relation

$$\frac{D}{L} = \frac{C_D}{C_L}$$

$$\dot{v} = -D/m = -\frac{C_D}{C_L m} \cdot L = -\frac{C_D}{C_L} \left(g_0 - \frac{v^2}{r_0} \right)$$

Answer: \textbf{Answer:}

$$a = -\frac{C_D}{C_L} \left(g_0 - \frac{v^2}{r_0} \right)$$

C4

0.50 pts

Let the initial speed of the spacecraft be $v_1 < v_s$ and the final speed $v_2 < v_1$. Find the horizontal displacement of the ship s during the motion. Consider that the entire change in the velocity module occurs due to the force of air resistance. Express your answer using r_0 , C_L , C_D , v_s , v_1 , v_2 .

Using the relations $a = dv/dt$ and $v = ds/dt$ (in the approximation $\cos \gamma \approx 1$), we obtain the connection between displacement and change in speed,

$$\frac{ds}{dv} = \frac{v}{a} = -\frac{C_L}{C_D} \frac{v}{g_0 - v^2/r_0} = -\frac{C_L}{C_D} r_0 \frac{v}{v_s^2 - v^2}.$$

Integrating, we get

$$s = -\frac{C_L}{C_D} r_0 \int_{v_1}^{v_2} \frac{v dv}{v_s^2 - v^2} = -\frac{C_L}{2C_D} r_0 \int_{v_1}^{v_2} dv \left(\frac{1}{v_s - v} + \frac{1}{v_s + v} \right) = \frac{C_L}{2C_D} r_0 \ln \frac{v_s^2 - v_2^2}{v_s^2 - v_1^2}.$$

Answer: \textbf{Answer:}

$$s = \frac{C_L}{2C_D} r_0 \ln \frac{v_s^2 - v_2^2}{v_s^2 - v_1^2}$$

C5

0.60 pts

Let the following parameters be given for the spaceship: $m = 84 \times 10^3 \text{ kg}$ $S = 250 \text{ m}^2$ $C_D = 0.8$ $C_L = 0.9$ Find the values of speed (in km/s) and acceleration due to air resistance (in units of g_0) at altitudes $h_1 = 80 \text{ km}$, $h_1 = 60 \text{ km}$, $h_1 = 40 \text{ km}$.

Find the values of speed (in km/s) and acceleration due to air resistance (in units of g_0) at heights $h_1 = 80 \text{ km}$, $h_1 = 60 \text{ km}$, $h_1 = 40 \text{ km}$.

Using formulas for speed and acceleration expressed in terms of speed, we obtain

Answer: \textbf{Answer:} $h, \text{ km}$ a/g_0 $v, \text{ km/c}$ 80 −0.13 7.3 60 −0.65 4.1 40 −0.87 1.2

C6

0.20 pts

Under the conditions of the previous paragraph, calculate the range of horizontal movement of the spacecraft when lowering from height $h_0 = 90 \text{ km}$ to height $h_f = 30 \text{ km}$.

Let's calculate the initial and final speeds of the spacecraft

$$v_1 = v(h_0) = 7.7 /, \quad v_2 = v(h_f) = 0.61 \text{ km/c},$$

then the range of movement

Answer: \textbf{Answer:}

$$s = 11.4 \cdot 10^3$$

D1

0.40 pts

Write down expressions for energy E and angular momentum L . Express your answer using G, m, a, e, M . Note: It may be convenient to write the ZSE at the apocentric or periapsis point.

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Let the distances from the ground to the ship at the pericenter (the closest point) and apocenter (the most distant point) be equal to r_1 and r_2 , respectively. The speeds of the ship at these points are $L/(mr_1)$ and $L/(mr_2)$, respectively. Then:

$$E = \frac{L^2}{2mr_1^2} - \frac{GMm}{r_1} = \frac{L^2}{2mr_2^2} - \frac{GMm}{r_2},$$

Let us express hence angular momentum L :

$$L^2 = 2mr_1^2 E_1 + 2GMm^2 r_1 = 2mr_2^2 E_2 + 2GMm^2 r_2.$$

Hence let us express энергию

$$E(r_2^2 - r_1^2) = GMm^2(r_1 - r_2),$$

and therefore

$$E = -\frac{GMm}{r_1 + r_2} = -\frac{GMm}{2a}.$$

Таким же обсом let us express angular momentum

$$\frac{L^2}{2m} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right) = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right), \quad L^2 = 2GMm^2 \frac{r_1 r_2}{r_1 + r_2}.$$

Expressing расстояния via/through большую полуось and эксцентриситет $r_1 = a(1 - e)$, $r_2 = a(1 + e)$ let us find

$$L^2 = GMm^2 a(1 - e^2).$$

Answer: \textbf{Answer:}

$$E = -\frac{GMm}{2a}, \quad L = \sqrt{GMm^2 a(1 - e^2)}$$

D2

0.20 pts

Write down the moment equation and expression for the power of the friction force P . Express your answer using $\vec{r}, \vec{v}, \alpha$.

Answer: \textbf{Answer:}

$$\frac{d}{dt} \vec{L} = -\alpha [\vec{r}, \vec{v}]$$

$$P = -\alpha v^2$$

D3

0.20 pts

Obtain an expression for the dependence of $\vec{L}$ on t if at time $t = 0$ $\vec{L} = \vec{L}_0$. Express your answer using $\alpha, m, \vec{L}_0$.

Let's use the definition for $\vec{L}$ and write the moment equation in the form

$$\frac{d}{dt} \vec{L} = -\frac{\alpha}{m} \vec{L},$$

Answer: \textbf{Answer:}

$$\vec{L} = \vec{L}_0 \exp\left(-\frac{\alpha}{m} t\right)$$

D4

0.30 pts

Show that:

$$[\vec{a}, \vec{L}] = \beta \frac{d}{dt} \vec{e}_i$$

Where $\vec{a}$ - acceleration ракеты, $\vec{e}_i$ - некоторый единичный vector полярных координат с началом в центре Земли. Find $\vec{e}_i$ and β . For/Atмечание: Может быть удобным использовать expression $\vec{L} = mr^2 \vec{\omega}$.

Note: It may be convenient to use the expression $\vec{L} = mr^2 \vec{\omega}$.
Newton's second law is

$$m\vec{a} = -\frac{GmM}{r^2} \vec{e}_r.$$

Substituting acceleration в vectorное произведение:

$$[\vec{a}, \vec{L}] = -\left[\frac{GM}{r^2} \vec{e}_r, \vec{L}\right] = -GM \left[\frac{1}{r^2} \vec{e}_r, mr^2 \vec{\omega}\right] = GmM [\vec{\omega}, \vec{e}_r] = GmM \frac{d}{dt} \vec{e}_r.$$

Answer: \textbf{Answer:}

$$\beta = GMm, \quad \vec{e}_i = \vec{e}_r$$

D5

0.30 pts

Multiplying expression (1) by $\vec{r}$ scalarly and using the results of point D4, obtain the dependence of the modulus of the radius vector $\vec{r}$ on φ – the angle between vectors $\vec{A}$ and $\vec{e}_r$. Express your answer using L, m, G, e, φ, M .

Multiply the expression by $\vec{r}$, we get

$$(\vec{r}, [\vec{v}, \vec{L}]) = GMmr + \vec{A} \cdot \vec{r}.$$

В левой части переставим множители, using свойства смешанного произведения:

$$(\vec{r}, [\vec{v}, \vec{L}]) = (\vec{L}, [\vec{r}, \vec{v}]) = \frac{L^2}{m}.$$

Then we get

$$\frac{L^2}{m} = GmMr(1 + e \cos \varphi),$$

where we find

Answer: \textbf{Answer:}

$$r = \frac{L^2/GMm^2}{1 + e \cos \varphi}$$

D6

0.50 pts

Using expression (1) and the results of point D4, obtain the dependence of the square of the speed v^2 on φ . Express your answer using L, G, m, e, φ, M .

Let's square expression (1):

$$[\vec{v}, \vec{L}]^2 = v^2 L^2 = (GmM)^2 + 2GmM\vec{e}_r \cdot \vec{A} + A^2 = (GmM)^2(1 + 2e \cos \varphi + e^2),$$

Answer: \textbf{Answer:}

$$v^2 = \frac{(GMm)^2}{L^2}(1 + e^2 + 2e \cos \varphi)$$

D7

0.40 pts

Obtain an expression for the time-average squared velocity $\langle v^2 \rangle$ as an integral over the angle φ . Express your answer in terms of $L, m, e, M, G, a, \varphi, \tau$, the rocket's orbital period. Note: It may be convenient to express the square of the average velocity from the averaged ZSE ($\langle T + U \rangle = \langle E \rangle$, where T is kinetic energy, U is potential).

Note: It may be convenient to express the square of the average velocity from the averaged ZSE ($\langle T + U \rangle = \langle E \rangle$, where T is kinetic energy, U is potential energy).

Method 1: By definition, the mean square of velocity is expressed through the integral as (τ – period of motion)

$$\langle v^2 \rangle = \frac{1}{\tau} \int_0^\tau v^2 dt.$$

Дифференциалы времени and угла φ можно связать с помощью momenta momentuma:

$$L = mr^2 \frac{d\varphi}{dt}.$$

Перейдем к интегрированию по углу

$$\langle v^2 \rangle = \frac{1}{\tau} \int_0^\tau v^2 \frac{dt}{d\varphi} d\varphi = \frac{m}{\tau L} \int_0^{2\pi} v^2 r^2 d\varphi.$$